

ORIGINAL RESEARCH

CORONARY

Performance of Large Language Models on the Acute Coronary Syndrome Guidelines Using Retrieval-Augmented Generation



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ABSTRACT

BACKGROUND Large language models (LLMs) are increasingly applied in interventional cardiology, but hallucinations limit their clinical utility.

OBJECTIVES The aim of this study was to assess whether retrieval-augmented generation (RAG), a technique that allows LLMs to access guideline content during response generation, improves accuracy when answering questions on the basis of the guidelines for acute coronary syndromes.

METHODS The accuracy of ChatGPT-4o, DeepSeek R1, and Med-PaLM 2 was compared using a set of 38 open-ended cardiology guideline-based questions and answers. ChatGPT-4o and DeepSeek R1 were evaluated with and without RAG, while Med-PaLM 2, a medicine-specific LLM, was tested without RAG. Model outputs were compared against guideline recommendations using an artificial intelligence-powered similarity score tool.

RESULTS DeepSeek R1 with RAG achieved the highest accuracy (94.7%; 95% CI: 82.7%-98.5%), followed by ChatGPT-4o with RAG (92.1%; 95% CI: 79.2%-97.3%) ($P = 0.922$). ChatGPT-4o without RAG achieved 71.1% accuracy (95% CI: 55.2%-83.0%), which significantly improved with RAG ($P = 0.017$). Among non-RAG models, DeepSeek R1 demonstrated the highest accuracy (78.9%; 95% CI: 63.7%-88.9%), followed by ChatGPT-4o without RAG (71.1%) ($P = 0.083$). Med-PaLM 2 had the lowest accuracy (68.4%; 95% CI: 52.5%-80.9%). Spearman correlation analysis revealed a strong correlation between DeepSeek R1 without RAG and Med-PaLM 2 ($r = 0.646$; 95% CI: 0.411-0.800; $P < 0.001$), indicating similar response patterns. Scatterplot analysis further revealed that RAG disproportionately improved lower scoring questions in DeepSeek R1 while improving scores more evenly in ChatGPT-4o.

CONCLUSIONS Embedding guideline content into LLM workflows via RAG can enhance LLM accuracy for clinical applications, particularly in scenarios common to interventional cardiology. These results support the potential for LLMs, when enhanced with domain-specific knowledge, to optimize clinical decision making and increase alignment with guidelines. (JACC Cardiovasc Interv. 2025;18:2458-2467) © 2025 by the American College of Cardiology Foundation.

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Large language models (LLMs) have gained significant recognition and are rapidly becoming indispensable tools in clinical research and practice. ChatGPT, launched by Open AI in November 2022, has 300 million weekly active users and processes more than 1 billion queries per day as of December 2024.¹ The health care industry, especially cardiology, has begun exploring the potential of LLMs to enhance clinical decision making.² Most cardiologists believe that artificial intelligence (AI) can increase diagnostic efficiency; in a recent survey of 521 interventional cardiologists, 60.6% anticipated the implementation of AI in their personal clinical practice within 5 years.^{3,4} However, challenges such as hallucinations—instances in which a model generates false or fabricated information despite sounding plausible—and factual inaccuracies remain significant barriers to wide clinical application.²

The role of LLMs in supporting clinical decision making in cardiology is still evolving, and physicians remain skeptical about their ability to provide up-to-date information and solutions for complex clinical scenarios.⁵ One proposed way to improve LLM accuracy and maintain their relevance is retrieval-augmented generation (RAG).⁶ RAG is a method that combines LLM with an external knowledge base, allowing real-time access to relevant documents during response generation.⁷ RAG comprises 2 key components: a retriever that locates relevant documents and a generator that synthesizes these documents into coherent outputs.^{8,9} It enhances the factual accuracy and domain specificity of model outputs, making it particularly valuable in medical contexts.¹⁰ RAG could enhance LLM performance by retrieving relevant information from an external document library during the generation process, grounding responses in factual data to improve accuracy, and provide up-to-date, domain-specific knowledge, especially in knowledge-intensive contexts.

Beyond ChatGPT, other LLMs, such as DeepSeek R1 (DeepSeek AI) and Med-PaLM 2 (Google DeepMind), are also being explored. DeepSeek has shown promise, but current evidence is extremely limited. As of February 12, 2025, a PubMed search for “DeepSeek” provided only 7 results, of which only 2 were research papers.^{11,12} The technical report for DeepSeek R1 (the latest model) demonstrated that it

achieved outstanding performance, particularly on reasoning tasks, comparable with leading closed-source models, while having the major advantage of training efficiency, significantly reducing computational costs while maintaining performance stability.¹³

Open-source LLMs have the advantages of higher accessibility and customization. However, closed-source LLMs, which include ChatGPT-4o and Med-PaLM 2, have historically outperformed open-source LLMs.¹⁴ Med-PaLM 2, developed by Google, is a medicine-specific LLM, currently available to a select group of Google Cloud customers for limited testing. Despite its restricted access, early research has shown promising results.¹⁵

METHODS

We aimed to compare the accuracy of different LLMs in answering open-ended questions on cardiology guidelines, specifically assessing the impact of RAG on performance (**Central Illustration**). We curated a set of 38 open-ended questions and answers from the companion document of the 2020 European Society of Cardiology (ESC) guidelines for managing acute coronary syndromes (ACS) in patients without persistent ST-segment elevation.¹⁶ These questions were specifically developed to complement the guideline content and reflect core diagnostic and risk stratification principles. Their use ensures alignment with guideline recommendations and reduces the potential for subjective bias from independently generating or selecting questions not formally linked to the guidelines.

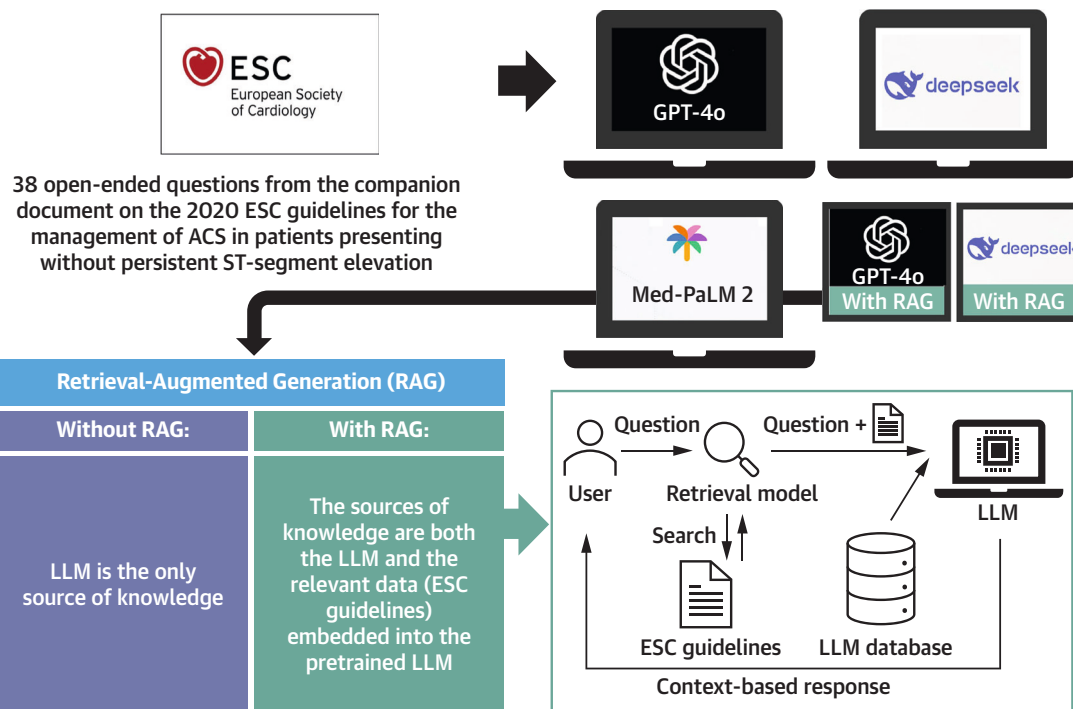
We tested ChatGPT-4o (Omni) and DeepSeek R1 by asking these questions directly from the document (without RAG) and again with the ESC guidelines embedded into a RAG model.^{17,18} Additionally, we evaluated Med-PaLM 2, testing its responses without RAG, as it is medicine specific.¹⁹ To access Med-PaLM 2, we obtained access for research purposes, as there is currently limited access to a specific group of users. To minimize variability, each model was prompted once per question, and the resulting single response was used for similarity scoring and accuracy assessment. The same prompt structure

ABBREVIATIONS AND ACRONYMS

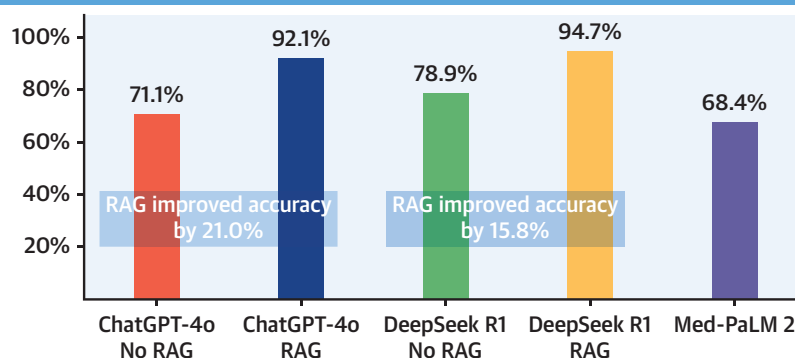
ACS	= acute coronary syndrome(s)
AI	= artificial intelligence
ESC	= European Society of Cardiology
LLM	= large language model
RAG	= retrieval-augmented generation

CENTRAL ILLUSTRATION Performance of LLMs on the ACS Guidelines Using RAG

Performance of Large Language Models on the Acute Coronary Syndrome Guidelines Using Retrieval-Augmented Generation



Accuracy



- Embedding the guidelines in the model using RAG increased Deep Seek R1's accuracy from 78.9% to 94.7%.
- The accuracy of ChatGPT-4o increased from 71.1% to 92.1% with RAG.
- The results support the potential for LLMs, when enhanced with domain-specific knowledge, to optimize clinical decision-making and increase alignment with the guidelines.
- However, AI should augment rather than replace clinical judgment.

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ACS = acute coronary syndrome(s); ESC = European Society of Cardiology; LLM = large language model; RAG = retrieval-augmented generation.

was used across all models to ensure consistency in input formatting and minimize prompt-induced variability (Supplemental Table 1).

We tried to include Meditron (Meta AI), a suite of open-source LLMs tailored to the medical field that was built on Meta Llama 3, in our comparative analysis, but it performed poorly and did not give relevant answers to the questions asked and was not included in the final analysis.²⁰

For our RAG analysis, Google Colab Notebook Python 3 version 3.11 was used. All models were asked the ESC guideline questions using the method of zero-shot prompting. To create a RAG model using ChatGPT 4o, the ESC guidelines PDF was first uploaded. Using the NLTK package, text from the PDF was split into chunks of 500 tokens. Sentence transformer using the model all-MiniLM-L6-v2 was used to create embeddings from the chunked tokens.^{21,22} A Facebook AI Similarity Search index was used to create vectors from the generated embeddings. In an OpenAI environment, the query vector retrieved the top 5 vectorized chunks on the basis of similarity score to give a response. The temperature of the model was set at 1, and the number of tokens generated was set at 2,048.

Similarly, to create a RAG model using DeepSeek R1, the langchain community package was installed.¹³ Using the recursive character text splitter function, the text from the PDF was split into chunks of 500 tokens. Sentence transformer using the model all-MiniLM-L6-v2 was used to create embeddings from the chunked tokens. A Facebook AI Similarity Search index was used to create vectors from the generated embeddings.²³ In a HuggingFace environment, the query vector retrieved the top 5 vectorized chunks on the basis of similarity score to give a response. The temperature of the model was set at 1, and the number of tokens generated was set at 2,048.

DeepEval version 2.3.7 was used to conduct a similarity analysis among the actual output, the output provided by the LLM and the expected output, and the recommendations from the ESC guidelines. It is an open-source LLM evaluation framework. We used the GEval function using an OpenAI environment to assess outputs using the prompt “Similarity - determine if the actual output has a similar core recommendation to the expected output. Ignore structural differences and details of other options.” Descriptive statistics, including accuracy percentages, were calculated to assess the overall performance of each model on the basis of the 0.5 similarity score threshold.

To evaluate the differences in similarity scores between models, we conducted a Friedman test and

TABLE 1 Key Performance Metrics for Each Model			
Model	Accuracy, %	Median Similarity (Q1-Q3)	Improvement With RAG, %
ChatGPT-4o (no RAG)	71.1	0.76 (0.43-0.90)	—
ChatGPT-4o (RAG)	92.1	0.86 (0.80-0.94)	+21.0
DeepSeek R1 (no RAG)	78.9	0.86 (0.68-0.95)	—
DeepSeek R1 (RAG)	94.7	0.88 (0.78-0.92)	+15.8
Med-PaLM 2	68.4	0.76 (0.32-0.89)	—
RAG = retrieval-augmented generation.			

Wilcoxon signed rank tests, assessing statistical significance with a threshold of $P \leq 0.05$. Spearman correlation analysis was used to assess relationships between similarity scores across models. We also computed median differences to provide insight into the distribution of similarity scores. Correlation strength was categorized as very weak ($r < 0.20$), weak ($0.20 \leq r < 0.40$), moderate ($0.40 \leq r < 0.60$), strong ($0.60 \leq r < 0.80$), and very strong ($r \geq 0.80$).²⁴

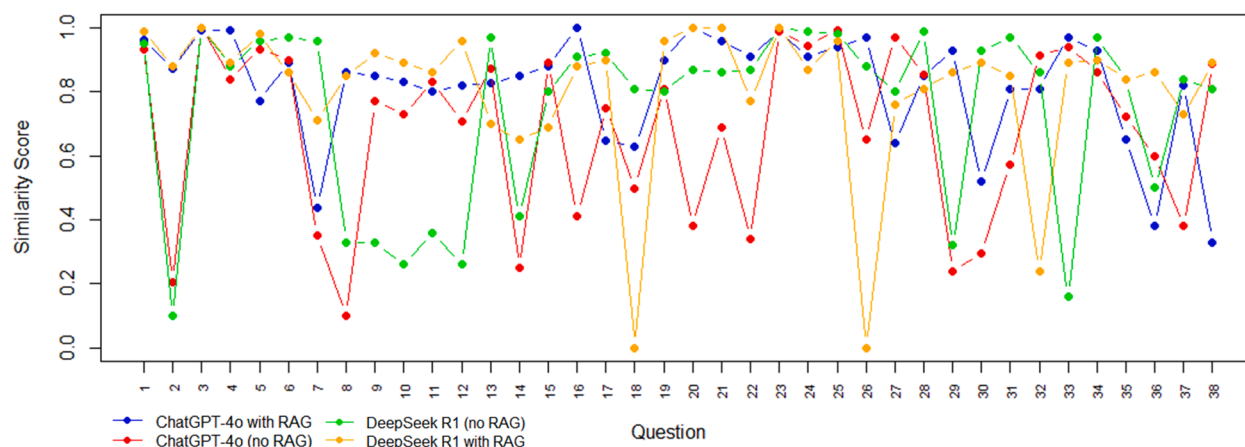
We performed these statistical analyses using both R version 4.4 (R Foundation for Statistical Computing) and ChatGPT-4o to verify the consistency of the results. The results generated by ChatGPT-4o were consistent with those obtained using R, confirming the reliability of AI-assisted statistical analysis.²⁵ P values <0.05 were considered to indicate statistical significance.

RESULTS

DeepSeek R1 achieved the highest accuracy, which increased from 78.9% (30 of 38 questions) to 94.7% (95% CI: 82.7%-98.5%; 36 of 38 questions) after embedding the guidelines in the model using RAG (Table 1). For ChatGPT-4o, accuracy was 71.1% (95% CI: 55.2%-83.0%; 27 of 38 questions) when the guidelines were not embedded (without RAG) and increased to 92.1% (95% CI: 79.2%-97.3%; 35 of 38 questions) with RAG, while Med-PaLM 2 scored the lowest (68.4%; 95% CI: 52.5%-80.9%; 26 of 38 questions).

Figure 1 presents histograms of similarity scores for each LLM, showing the distributions of similarity scores and highlighting differences in performance and consistency. Comparing the performance of ChatGPT-4o and DeepSeek R1 before the use of RAG, there was a trend toward higher accuracy for DeepSeek R1, though it did not reach statistical significance ($P = 0.083$). The median similarity score was significantly higher for ChatGPT-4o with RAG (0.86 [Q1-Q3: 0.80-0.94] vs 0.76 [Q1-Q3: 0.32-0.89];

FIGURE 1 Paired Line Plot of ChatGPT-4o and DeepSeek R1 (RAG, No RAG)



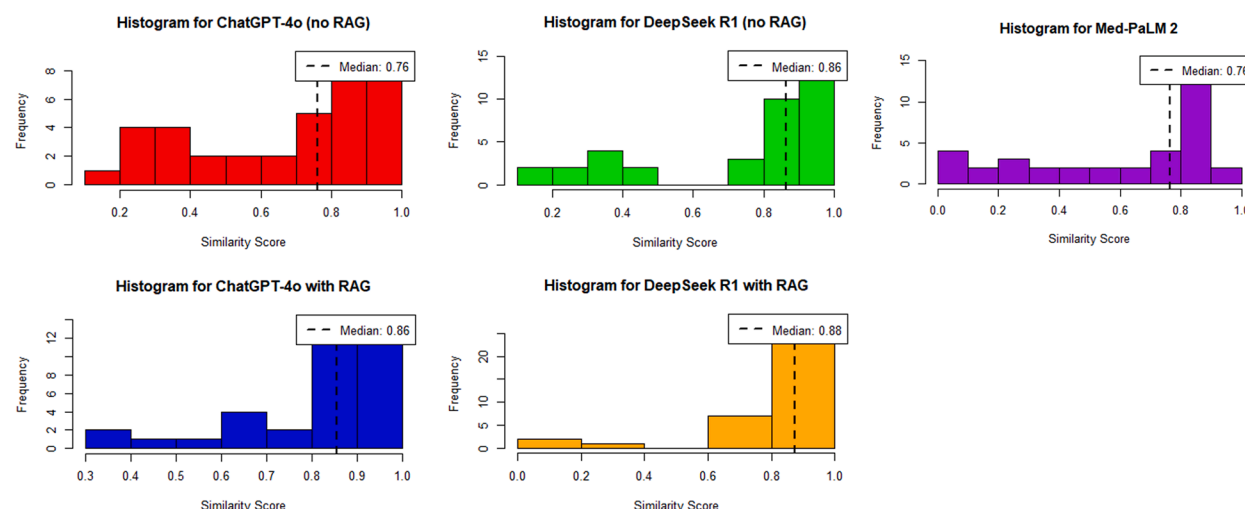
RAG = retrieval-augmented generation.

$P = 0.017$), DeepSeek R1 without RAG (0.86 [Q1-Q3: 0.68-0.95] vs 0.76 [Q1-Q3: 0.32-0.89]; $P < 0.001$), and DeepSeek R1 with RAG (0.88 [Q1-Q3: 0.78-0.92] vs 0.76 [Q1-Q3: 0.32-0.89]; $P = 0.003$) compared with Med-PaLM 2. No statistically significant differences were found when comparing the median similarity scores of ChatGPT-4o without RAG vs Med-PaLM 2

($P = 0.437$) and ChatGPT-4o with RAG vs DeepSeek R1 without RAG ($P = 0.596$).

A paired line plot for the similarity scores of the models that were enhanced with RAG for each question is shown in [Figure 2](#). No significant differences were found comparing the median similarity scores of the 2 LLMs that used RAG, ChatGPT-4o and

FIGURE 2 Histograms of Similarity Scores for Each Large Language Model



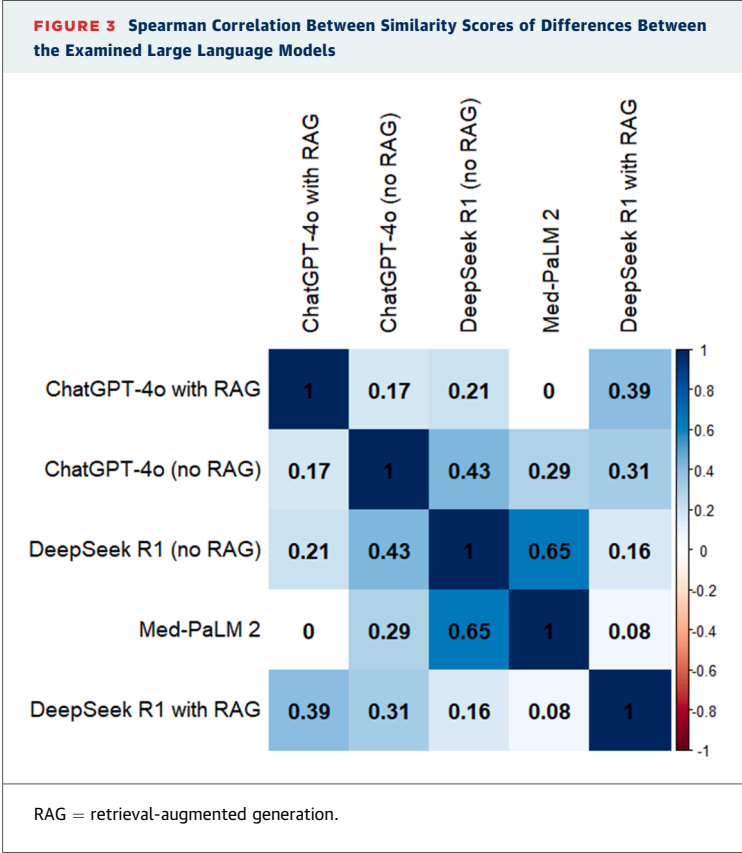
RAG = retrieval-augmented generation.

DeepSeek R1 ($P = 0.922$). The median similarity score was higher when the guidelines were embedded (RAG model) compared with when the guidelines were not embedded (0.86 [Q1-Q3: 0.80-0.94] vs 0.76 [Q1-Q3: 0.43-0.90]; $P = 0.017$) for ChatGPT-4o, while it was not statistically significant for DeepSeek R1 (0.88 [Q1-Q3: 0.78-0.92] vs 0.86 [Q1-Q3: 0.68-0.95]; $P = 0.544$).

One of the most notable differences in similarities scores when comparing the RAG vs no RAG model of ChatGPT-4o, was observed in question 2: “A 72-year-old patient with hypertension and hypercholesterolaemia as cardiovascular risk factors (CVRf) presents to the emergency department with typical chest pain of 3 h duration, palpitations, atrial fibrillation with a ventricular rate of about 120 beats per minute, ST-segment depression on electrocardiogram (ECG), and a mild elevation in cardiac troponin (cTn) (twice the upper limit of normal [ULN]). Is it correct to state that the underlying process is a rupture, ulceration, fissuring, or erosion of a coronary atherosclerotic plaque?” The guideline companion document’s answer was negative, clarifying that the process could be caused by an extracoronary condition causing imbalance between myocardial oxygen supply and demand (eg, tachycardia, anemia, hypertension, hypotension). Without RAG, the LLM incorrectly stated that the process was likely plaque related. With RAG, the response aligned with the guideline, accurately emphasizing the possibility of type 2 myocardial infarction and the need for further evidence to confirm plaque rupture. Examples illustrating how RAG improved model performance are provided in [Supplemental Table 2](#).

A Spearman correlation analysis was conducted to evaluate the relationships between the similarity scores of different LLMs ([Figure 3](#)). The strongest correlation was observed between DeepSeek R1 without RAG and Med-PaLM 2 ($r = 0.646$; 95% CI: 0.411-0.800; $P < 0.001$), indicating a high degree of similarity in their response pattern across different questions. Moderate correlation was found between ChatGPT-4o without RAG and DeepSeek R1 without RAG ($r = 0.429$; 95% CI: 0.126-0.658; $P = 0.007$) and weak between ChatGPT-4o with RAG and DeepSeek R1 with RAG ($r = 0.392$; 95% CI: 0.083-0.632; $P = 0.015$), suggesting some level of alignment between these models’ outputs.

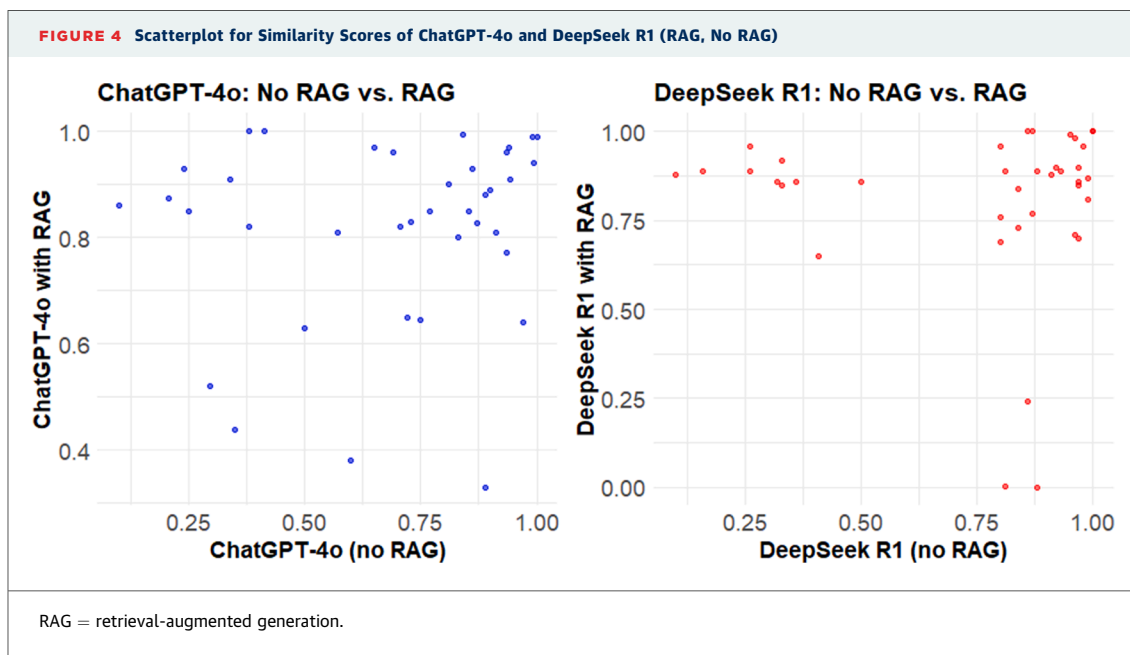
Conversely, very weak or nonsignificant correlations were identified between several models. DeepSeek R1 without RAG and DeepSeek R1 with RAG ($r = 0.157$; 95% CI: -0.171 to 0.454; $P = 0.346$) exhibited a very weak correlation. However, the scatterplot with the visual evidence indicates that



RAG significantly boosted similarity scores for questions that initially had lower scores without RAG ([Figure 4](#)). Unlike DeepSeek R1, the ChatGPT-4o scatterplot suggests that RAG’s influence was more balanced, with a very weak correlation between ChatGPT-4o with RAG and ChatGPT-4o without RAG ($r = 0.171$; 95% CI: -0.158 to 0.465; $P = 0.306$).

DISCUSSION

Our study demonstrated that the RAG model improved significantly the accuracy of ChatGPT-4o (from 71.1% to 92.1%), while for DeepSeek R1 (from 78.9% to 94.7%), the improvement was more apparent for questions that initially had lower scores. Among the models evaluated without RAG, DeepSeek R1 outperformed both ChatGPT-4o and Med-PaLM 2, achieving the highest accuracy (78.9%) in this setting. Although the difference between DeepSeek R1 and ChatGPT-4o without RAG was not statistically significant ($P = 0.083$), the trend suggests that DeepSeek R1 may operate better than ChatGPT-4o when no retrieval augmentation is used. In contrast, Med-PaLM 2 had the lowest accuracy (68.4%) despite being designed for medical use.



There has been a growing literature on ChatGPT’s performance in medicine and cardiology, and it has been outperforming other LLMs.^{26–28} However, evidence on DeepSeek is extremely limited, making this the first study examining the performance of DeepSeek R1 on cardiology-related questions. DeepSeek uses chain-of-thought reasoning, which increases explainability, creating a “thinking aloud” sensation for the user.²⁹ This could be particularly useful for the clinical environment, in which the reasoning is extremely important, as it can foster a collaborative AI-human interaction and increase the trust and adoption by clinicians.³⁰ Additionally, DeepSeek R1 is an open-source LLM, with all the inherent advantages of such models, including the democratization of innovation and higher accessibility.^{14,31}

Although DeepSeek R1 demonstrated strong accuracy, its adoption in the United States faces significant regulatory challenges. As an AI model developed by a China-based startup, it is currently subject to data security concerns and restrictions from U.S. federal and state officials, limiting its use in government, health care and other infrastructure sections.³² In contrast, the medicine-specific LLM, Med-PaLM 2, demonstrated the lowest performance, suggesting that factors such as model architecture, training data, and optimization strategies play a more significant role in clinical question-answering accuracy.

In our study, RAG improved the performance of the LLMs examined, allowing them to achieve excellent scores. There are some early results that

RAG can improve accuracy and clinical reasoning. A computational study including 7,663 questions in biomedicine demonstrated that RAG improved the accuracy of 6 different LLMs by up to 18% over chain-of-thought prompting.³³ In the field of orthopedics, RAG improved the accuracy of every LLM by an average 39.7%, when examined on the performance of 100 questions and answers based on the 2022 American Academy of Orthopaedic Surgeons anterior cruciate ligament guidelines.³⁴ Similar results have been observed in hepatology, ophthalmology, and oncology.^{35–37} To further contextualize our findings, a recent study evaluating ChatGPT-4 on the American College of Cardiology/Society for Cardiovascular Angiography and Interventions interventional cardiology board-style examination reported an overall accuracy of 78%. In the same study, interventional cardiology fellows achieved an average accuracy of 82.2%, outperforming ChatGPT-4.³⁸ In contrast, our study demonstrated that when guideline content is embedded via RAG, ChatGPT-4o achieved 92.1% accuracy, and DeepSeek R1 achieved 94.7% accuracy, underscoring the importance of retrieval augmentation for high-fidelity performance in cardiology-related clinical decision making. However, the content and format of the questions differed between the 2 studies.

An additional advantage of RAG is bias reduction and disparity mitigation.¹⁸ Ethical concerns have been repeatedly expressed by researchers regarding this inherent limitation of the LLMs.³⁹ They usually

lack diverse data sets, and there is evidence that this can result in selective accuracy.^{40,41} In a recent study, ChatGPT exhibited gender and racial biases in ACS management; when specifying patients as female, African American, or Hispanic, the researchers observed a decrease in guideline-recommended medical management, diagnosis, and symptom management of ACS.⁴⁰ However, whether RAG meaningfully mitigates these disparities needs to be directly tested. Future research is needed to determine if embedding guideline-based content can systematically improve fairness across patient subgroups. Future research could also explore the use of counterfactual clinical scenarios to assess whether RAG-enhanced LLMs maintain fairness across patient subgroups. Such testing would be critical to understanding the role of RAG in mitigating existing biases in AI-generated clinical recommendations.

Our results suggest that developing and deploying a customized cardiology-trained LLM may help overcome some of the current LLM limitations. Pretraining LLMs on cardiology specific knowledge can optimize the use of these models in real-world settings and ensure a high level of accuracy. For clinicians, the most immediate implication of this study is the potential to use retrieval-augmented LLMs to support evidence-based decisions at the point of care, particularly in complex cases in which guideline nuances matter. For example, incorporating RAG-enhanced tools into electronic health record systems could allow physicians to ask natural language questions and receive guideline-aligned recommendations in real time. This could be especially valuable in acute settings such as the emergency department or cardiac catheterization laboratory, where time-sensitive decisions are critical.

The idea of domain-specific LLMs is not new. Future development on medical LLMs should focus of building these domain-specific LLMs on top of high-performing, general-purpose architectures, incorporating state-of-the-art reasoning, retrieval mechanisms, and training optimizations to ensure both accuracy and adaptability in real-world clinical applications. However, cardiologists must still interpret these with critical thinking, as hallucinations still occur, and accuracy is not perfect.^{5,38,42} This highlights the importance of fostering collaboration between human expertise and AI systems in health care, ensuring that AI augments rather than replaces clinical judgment.

STUDY LIMITATIONS. The number of questions was limited ($n = 38$). Although this companion set provides a strong foundation rooted in guideline-aligned principles, we acknowledge that it may not encompass all clinical scenarios encountered in ACS care. Future work should aim to expand the test set to include additional domains.

Med-PaLM 2 was evaluated without RAG enhancement because of access restrictions not allowing integration with external document retrieval workflows. As such, its performance in this study should be interpreted cautiously and not directly compared with the RAG-enhanced ChatGPT-4o and DeepSeek R1 models. The observed differences likely reflect, at least in part, the absence of contextual augmentation rather than inherent limitations of the Med-PaLM 2 architecture. We therefore present the Med-PaLM 2 findings as exploratory.

Another limitation is the variability in responses generated by the LLM, which can affect similarity scores. As newer models are developed, the performance of these models may differ, potentially leading to improved or worse results.⁵

CONCLUSIONS

Our study shows that DeepSeek R1 and ChatGPT-4o perform better in answering open-ended cardiology questions when the relevant guidelines are embedded in a RAG system. Embedding domain-specific knowledge, such as clinical guidelines, in LLMs could address some current limitations and significantly enhance their ability to support clinical decision making. Future research should focus on optimizing LLMs for cardiology-specific applications to ensure maximum accuracy, transparency, and fairness.

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PERSPECTIVES

WHAT IS KNOWN? LLMs such as ChatGPT are being explored for medical applications but concerns about hallucinations and inaccurate responses have limited their clinical adoption.

WHAT IS NEW? This study demonstrates that integrating guideline documents into LLM workflows using RAG significantly improves the accuracy of responses to complex clinical questions in the field of interventional cardiology.

WHAT IS NEXT? Development of cardiology-specific LLMs trained on interventional data sets and decision pathways could potentially facilitate clinician decision making. Future work should focus on integrating these models into clinical environments in a safe, transparent, and regulation-compliant manner to support care teams in real time.

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APPENDIX For supplemental tables, please see the online version of this paper.